Mitigating Inequality and Manipulation in Multi-Object Allocation

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Abstract

In multi-object assignment problems without monetary transfers—such as course allocation and task assignments—sequential mechanisms generate a tension between incentives and fairness. Allowing agents to select several objects in priority order is strategy-proof but can concentrate too much value on those who choose first. Distributing choices across rounds reduces this advantage, but predictable future turns create opportunities for manipulation. We introduce the Switching Random Priority mechanism (SRP), which randomly selects each successive picker among agents other than the previous picker. This prevents consecutive choices while making it risky to rank a desirable object lower. We show that, with at least three agents, SRP is weakly SD-strategy-proof and weakly SD-envy-free. Moreover, every realized allocation satisfies ordinal rank balance, a fairness property limiting the concentration of highly ranked objects that quota-based serial dictatorship does not generally satisfy. With two agents, SRP instead becomes SD-efficient but fails weak SD-strategy-proofness.

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1 Introduction

The allocation of indivisible goods among agents without monetary transfers is a central problem in market design, spanning domains such as course allocation, athlete assignments to sports teams, and task assignment. When agents demand multiple objects rather than a single item, designing a mechanism that balances strategy-proofness, fairness, and efficiency becomes more complex.

A natural starting point is random serial dictatorship (RSD): agents are ordered uniformly at random and choose from the remaining objects according to that order. With no restriction on how many objects an agent may receive, and when all objects are desirable, the first agent takes everything. The rule is strategy-proof and symmetric before the priority order is drawn, but its realized allocations are maximally unequal in quantity and welfare. One response to this issue is to impose quotas. Under quota-RSD, agents are randomly ordered, and each consumes her entire quota by sequentially selecting the most preferred remaining objects. This retains strategy-proofness under responsive preferences and guarantees equal numbers of objects, but it need not equalize their quality: an early agent may obtain many widely preferred objects before later agents make any selection. Quotas therefore address one dimension of ex-post inequality while leaving the advantage of priority largely intact (Zhou, 1990; Bogomolnaia and Moulin, 2001; Kojima, 2009; Bouveret *et al.*, 2025).

Draft mechanisms take a different approach. Rather than allowing each agent to exhaust her quota in one block, they distribute picking opportunities over successive rounds. In a cyclic draft, a priority order is repeated; in a snake draft, the order reverses from one round to the next. These procedures keep bundle sizes equal, or nearly so, while preventing any agent from having an extended initial monopoly over the available objects. They also retain a natural ex-post efficiency guarantee: there is no reallocation that replaces every object in every agent's bundle with a distinct, weakly preferred object. Their weakness is incentives. An agent who knows when she will pick again has an incentive to rank a contested object ahead of a personally valuable but uncontested one, expecting to collect the deferred object in a later round. Thus, the predictability that helps drafts distribute opportunities also undermines strategy-proofness (Brams and Straffin, 1979; Budish and Cantillon, 2012; Bouveret and Lang, 2014; Coreno and Balbuzanov, 2026).

This paper studies whether randomizing the identity of the next picker can improve the incentives of a draft without abandoning sequential choice. We introduce the *Switching Random Priority* (SRP) mechanism, a centralized allocation rule based on agents' reported rankings of individual objects. The mechanism selects the first agent uniformly at random and, after each object is assigned, selects the next agent uniformly among those other than

the current picker. The selected agent always receives her highest-ranked remaining object. The resulting Markov chain never assigns consecutive turns to the same agent, but—unlike a cyclic or snake draft—does not guarantee when, or even whether, an agent will receive another turn. Consequently, ranking a desirable object lower entails a risk that is absent when the agent's future turns are predetermined.

Our first result formalizes this incentive intuition. For at least three agents, SRP is *weakly SD-strategy-proof*: no misreport yields an assignment that strictly improves on truthful reporting by stochastic dominance. We next establish SRP's fairness and efficiency properties, distinguishing guarantees about the lottery from guarantees about each realized allocation. For every number of agents, SRP is anonymous, treats equals equally, and is weakly SD-envy-free: no agent's probabilistic assignment is strictly SD-dominated by another agent's assignment from her own perspective. Every realized allocation is also *pairwise-dominance efficient*, a property shared by all benchmark mechanisms. More importantly, for the distribution of picking opportunities, every realization satisfies *ordinal rank balance*: relative to any object an agent receives, the number of better objects in her own bundle cannot exceed the number of objects held by other agents that their respective recipients rank above that object. Thus, the switching requirement yields a restriction on the concentration of highly ranked assignments even after the random turns have been realized. SRP does not, however, attain SD-efficiency in general when there are at least three agents and three objects.

The two-agent case presents some substantive differences. Once the first picker has been drawn, switching forces strict alternation. The uncertainty over future turns that supports weak SD-strategy-proofness with three or more agents disappears, and the mechanism can be manipulated by a strict SD improvement. However, and perhaps surprisingly, with two agents SRP is SD-efficient. When the number of objects is even, its realized allocations additionally satisfy one-object unidirectional and bidirectional envy bounds under responsive preferences.

Finally, we examine inequality in realized outcomes. Symmetry ensures that every agent receives the same expected number of objects, although realized bundle sizes need not be identical. To study inequality in the value of allocated bundles, we derive a closed-form expression for expected across-agent payoff variance when all agents have identical additive values drawn independently from a uniform distribution. This benchmark isolates the inequality generated by the sequence of picks, independently of disagreements about the ranking of objects.

1.1 Literature Review

The literature on indivisible object allocation treats serial dictatorship as a central benchmark, where agents are processed in a fixed priority order, with each being assigned her most preferred set of remaining objects. For single-object allocation, Svensson (1999) shows that strategy-proofness, non-bossiness, and neutrality together characterize serial dictatorship. When generalized to multiple objects, the mechanism is characterized by combining strategy-proofness and Pareto efficiency with either population-monotonicity (Klaus and Miyagawa, 2002) or non-bossiness together with either resource-monotonicity (Ehlers and Klaus, 2003) or neutrality under quota constraints (Hatfield, 2009). While strategy-proof and Pareto optimal under the relevant feasibility constraints, serial dictatorship can produce substantial ex-post inequality in multi-object settings. To ensure ex-ante fairness, the literature commonly randomizes the priority order, leading to Random Serial Dictatorship (RSD). Quota-RSD addresses inequality in the number of objects received, but it does not eliminate the advantage of earlier agents in the desirability of their assignments (Kojima, 2009). Bouveret *et al.* (2025) study how unequal quotas can compensate for this priority advantage.

Draft mechanisms rotate the picking turn among agents rather than allowing each agent to select her entire quota in consecutive turns. Brams and Straffin (1979) establish a pairwise efficiency property of sincere picking, which we formulate below as pairwise-dominance efficiency, although draft mechanisms are not strategy-proof. Coreno and Balbuzanov (2026) provide an axiomatic characterization of draft rules and establish impossibility results concerning the compatibility of their fairness properties with strategy-proofness. In contrast, the booster draft mechanism (Caspari, 2026) ensures strategy-proofness by decomposing the allocation into independent picking sequences over disjoint subsets of objects, but at the cost of Pareto efficiency. Empirically, Budish and Cantillon (2012) provide evidence from Harvard's course allocation draft, where a random priority draw initializes the sequence to provide ex-ante symmetry. They document that the draft yields higher aggregate welfare than a random serial dictatorship benchmark adapted to the course-allocation environment, despite being manipulable.

Several fairness concepts have been proposed in attempts to approximate envy-freeness. Amanatidis *et al.* (2017) demonstrate that no deterministic strategy-proof mechanism allocating all objects guarantees envy-freeness up to one good (EF1) for two agents with additive valuations when there are at least five objects. Other fairness notions include EFX (envy-free up to any item) (Caragiannis *et al.*, 2019) and maximin share (MMS) fairness (Kurokawa *et al.*, 2018). Working under preferences satisfying the dominance relation, Caspari (2026) introduces a bidirectional measure of bounded envy-freeness. Our analysis adopts this distinction to establish bounded envy guarantees for the two-agent case under responsive

preferences. For arbitrary numbers of agents, we introduce ordinal rank balance, a property restricting the concentration of highly ranked objects within each realized allocation.

A separate literature explores probabilistic assignment mechanisms to achieve ex-ante symmetry, yet encounters fundamental impossibility results. In the unit-demand setting, Bogomolnaia and Moulin (2001) document a fundamental trade-off: Random Serial Dictatorship is strategy-proof and weakly envy-free but fails SD-efficiency, whereas the Probabilistic Serial mechanism achieves SD-efficiency and envy-freeness at the cost of full strategy-proofness, while retaining weak SD-strategy-proofness. They prove that no mechanism can jointly satisfy SD-efficiency, strategy-proofness, and equal treatment of equals for four or more agents. Subsequent work examines related incompatibilities in the presence of indifferences and on restricted preference domains (Katta and Sethuraman, 2006; Kasajima, 2013). In settings where agents demand multiple objects, Kojima (2009) shows that SD-efficiency, SD-envy-freeness, and weak SD-strategy-proofness are incompatible under fixed quotas and additively separable preferences, while Aziz and Kasajima (2017) establish analogous impossibilities under related combinations of axioms. Our mechanism retains weak SD-strategy-proofness for three or more agents by relaxing SD-efficiency and instead guaranteeing ex-post pairwise-dominance efficiency and ordinal rank balance.

The remainder of the paper proceeds as follows. Section 2 introduces the assignment problem. Section 3 proposes the Switching Random Priority mechanism and its benchmarks, and Section 4 establishes its theoretical properties. Section 5 examines the two-agent case. Section 6 derives the expected payoff variance of SRP.

2 Model

An assignment problem is a triple (N, O, P). $N = \{1, \ldots, n\}$ represents the set of agents, where $n \geq 2$.[2] The set of goods is denoted as $O = \{o_1, o_2, \ldots, o_m\}$, comprising $m \geq 2$ indivisible objects. We use lowercase letters for individual objects and uppercase letters for bundles (subsets of O); in particular, the singleton bundle $\{o\}$ is distinct from the object o. Each agent $i \in N$ has preferences over bundles $S \subseteq O$, represented by a reflexive and transitive weak relation R_i and its strict part P_i, with a strict and complete ranking of individual objects. Our general results use only this object ranking; additional assumptions on bundle preferences are stated where they are needed. We also use P_i and R_i for the induced ranking of individual objects: xP_iy means $\{x\}P_i\{y\}$, and similarly for R_i. Singleton bundles retain their braces. For any object $o \in O$, let $\mathrm{Up}_i(o) = \{o' \in O : o'R_io\}$ denote agent i's upper-contour set at o, consisting of the objects weakly preferred to o.

[2]Some results require $n \geq 3$, but state this restriction explicitly.

A deterministic allocation $A = (A_i)_{i \in N}$ consists of pairwise disjoint bundles $A_i \subseteq O$ such that $\bigcup_{i \in N} A_i = O$, where A_i is the bundle assigned to agent i. A probabilistic allocation is a matrix $\pi = [\pi_i(o_j)]_{i \in N, o_j \in O}$, where $\pi_i(o_j) \in [0,1]$ denotes the probability that object o_j is allocated to agent i. For every $o_j \in O$, $\sum_{i \in N} \pi_i(o_j) = 1$, so each object is allocated with probability one. Let Π be the set of all feasible probabilistic allocations. For each agent i, her probabilistic assignment in $\pi \in \Pi$ is the vector $\pi_i = [\pi_i(o_j)]_{o_j \in O}$. For any $S \subseteq O$, write $\pi_i(S) = \sum_{o \in S} \pi_i(o)$. Let $\mathcal{P}_i$ denote the set of strict linear orders over O, and let $\mathcal{P} = \prod_{i \in N} \mathcal{P}_i$. A preference profile $P = (P_i)_{i \in N} \in \mathcal{P}$ specifies the agents' rankings of individual objects. These rankings are the only inputs to the mechanism; preferences over bundles, when relevant, are assumed to be consistent with these rankings. A mechanism $\phi : \mathcal{P} \to \Pi$ maps each preference profile to a probabilistic allocation.

2.1 Desirable Properties

We evaluate probabilistic allocation mechanisms against properties of incentives, fairness, and efficiency under first-order stochastic dominance (SD). Consider two probabilistic allocations $\pi, \pi' \in \Pi$. We say π_i **weakly stochastically dominates** π'_i under preference P_i, denoted $\pi_i R_i^{\text{sd}} \pi'_i$, if

$$\sum_{o' \in \text{Up}_i(o)} \pi_i(o') \geq \sum_{o' \in \text{Up}_i(o)} \pi'_i(o') \qquad \text{for every } o \in O.$$

We say π_i **strictly stochastically dominates** π'_i, denoted $\pi_i P_i^{\text{sd}} \pi'_i$, if $\pi_i R_i^{\text{sd}} \pi'_i$ and the inequality is strict for at least one object $o \in O$. Since object rankings are strict, weak SD holds if and only if either the two assignment vectors are equal or strict SD holds.

Definition 2.1 (Weak SD-Strategy-Proofness). A mechanism ϕ is weakly SD-strategy-proof if for every agent $i \in N$, every preference profile $P \in \mathcal{P}$, and every misreport $P'_i \in \mathcal{P}_i$,

$$\begin{aligned} & \phi_i(P'_i, P_{-i}) R_i^{\text{sd}} \phi_i(P_i, P_{-i}) \\ \implies \quad & \phi_i(P'_i, P_{-i}) = \phi_i(P_i, P_{-i}). \end{aligned}$$

Equivalently, no misreport yields a strict SD improvement over truthful reporting.

Definition 2.2 (Weak SD-Envy-Freeness). An allocation π is weakly SD-envy-free at preference profile P if, for any $i \in N$, there is no $j \in N$ such that $\pi_j P_i^{\text{sd}} \pi_i$. A mechanism ϕ is weakly SD-envy-free if $\phi(P)$ is weakly SD-envy-free for all $P \in \mathcal{P}$.

Definition 2.3 (Anonymity). A mechanism ϕ is anonymous if for every permutation ρ of N and profile P,

$$\phi_{\rho(i)}(P^{\rho}) = \phi_i(P) \quad \text{for all } i \in N,$$

where $P^{\rho}_{\rho(i)} = P_i$.

Definition 2.4 (Equal Treatment of Equals). A mechanism ϕ satisfies equal treatment of equals if for all agents $i, j \in N$ and every preference profile $P \in \mathcal{P}$, when $P_i = P_j$, then the allocations must satisfy: $\phi_i(P) = \phi_j(P)$.

Definition 2.5 (SD-Efficiency). An allocation π is SD efficient at preference profile P if there exists no alternative allocation $\pi' \in \Pi$ such that $\pi'_i R_i^{\text{sd}} \pi_i$ for all $i \in N$, and $\pi'_i P_i^{\text{sd}} \pi_i$ for at least one $i \in N$. A mechanism ϕ is SD efficient if $\phi(P)$ is SD efficient for all $P \in \mathcal{P}$.

Definition 2.6 (Pairwise-dominance efficiency). An allocation A is *pairwise-dominance efficient* if there is no other feasible allocation $B \neq A$ such that, for every agent i, there exists an injection $f_i : A_i \to B_i$ satisfying

$$f_i(x) R_i x \qquad \text{for every } x \in A_i.$$

In words, an allocation is pairwise-dominance efficient if there is no reallocation in which every agent can replace each object she receives with a distinct object she weakly prefers. The property requires only rankings of individual objects, without imposing preferences over bundles.

3 The Switching Random Priority Mechanism and its counterparts

We now introduce the SRP mechanism. It operates as follows. After a uniformly random choice of who makes the initial draw, the picking turn always passes to a different agent, chosen uniformly from the other $n - 1$ agents.

Let $\sigma_t \in N$ denote the agent assigned to round t. Given the reported rankings of individual objects in P, the mechanism operates in m rounds: at each round t, agent σ_t receives her most preferred available object according to P_{σ_t}. The turn sequence $\sigma = (\sigma_1, \ldots, \sigma_m)$ is generated by a Markov process:

- **Initial draw:** The initial agent σ_1 is drawn uniformly at random from N, with $\Pr(\sigma_1 = i) = \frac{1}{n}$ for all $i \in N$.

- **Transition rule:** For each $t \geq 1$, the next active agent $\sigma_{t+1} \in N$ is determined by the $n \times n$ transition matrix T, defined as:

$$T_{ij} \equiv \Pr(\sigma_{t+1} = j \mid \sigma_t = i) = \begin{cases} 0 & \text{if } j = i, \\ \dfrac{1}{n-1} & \text{if } j \neq i. \end{cases} \tag{3.1}$$

Let $\Omega = \{\sigma \in N^m : \sigma_{t+1} \neq \sigma_t \text{ for } t = 1, \ldots, m-1\}$ denote the set of feasible picking sequences. Each $\sigma \in \Omega$ occurs with probability $1/[n(n-1)^{m-1}]$, independently of the reported preferences.

The following mechanisms will be treated as benchmarks to which we compare SRP.

Benchmark mechanisms. Each benchmark draws a permutation $\rho = (\rho_1, \ldots, \rho_n)$ of N uniformly at random, independently of P. At each round $t = 1, \ldots, m$, agent σ_t receives her most preferred available object according to P_{σ_t}, with the turn sequence defined as follows.

- **Quota random serial dictatorship (quota-RSD).** Assume $m = kn$ for a positive integer k and set $\sigma_t = \rho_{\lceil t/k \rceil}$. Each agent thus receives k objects in consecutive turns.
- **Cyclic draft.** Set $\sigma_t = \rho_r$, where $r = 1 + ((t-1) \bmod n)$. This repeats the order ρ until all objects are allocated.
- **Snake draft.** Write $t = qn + r$ with $q \geq 0$ and $1 \leq r \leq n$, and set

$$\sigma_t = \begin{cases} \rho_r, & q \text{ even}, \\ \rho_{n+1-r}, & q \text{ odd}. \end{cases}$$

The order reverses after every n turns, continuing until all objects are allocated.

4 Incentives, Fairness, and Efficiency

We now examine the normative properties of the mechanism under stochastic dominance, focusing on incentive compatibility, fairness, and efficiency.

We begin with strategy-proofness. The following example illustrates a profitable manipulation under cyclic draft.

Example 4.1 (Manipulating cyclic draft). There are three agents and six objects:

$$\begin{aligned} P_1 &: \quad a, b, c, d, e, f, \\ P_2 = P_3 &: \quad b, c, d, e, f, a. \end{aligned}$$

Agent 1 swaps a and b in her report. If she is first in the randomly drawn order (probability $1/3$), she receives $\{a, b\}$ instead of $\{a, d\}$; otherwise, her bundle is unchanged. The misreport therefore shifts marginal probability $1/3$ from d to b, strictly improving her assignment under SD.

The key is her guaranteed second turn: she can take b first and safely leave a, which both other agents rank last, for the next round.

The example shows the key intuition behind manipulations in draft mechanisms: Agent 1 knows she will be able to pick again before either 2 or 3 are able to pick object a. Therefore, it is safe for her to postpone it and prioritize b instead, which is also desired by the other players.

SRP removes this certainty. If agent 1 skips object a, there is a non-zero probability that she will not be able to get back to it later. This is the channel that makes SRP weakly SD-strategy-proof. With $n = 2$, this continuation is unavailable, and Proposition 5.1 shows that weak SD-strategy-proofness fails in general.

Theorem 4.2. *For $n \geq 3$, the SRP mechanism is weakly SD-strategy-proof.*

See Appendix A.1 for the proof.

Beyond incentive compatibility, the SRP mechanism provides fairness guarantees that hold for any number of agents.

Proposition 4.3. *SRP satisfies anonymity, equal treatment of equals, and weak SD-envy-freeness.*

See Appendix A.2 for the proof.

Having established the fairness and incentive properties of the mechanism, we now turn to its efficiency. To analyze SD-efficiency, we rely on a characterization based on the acyclicity of binary relations.

Lemma 4.4. *A random allocation π is SD-efficient if and only if the relation $\triangleright_{\pi,P}$ on O is acyclic, where*

$$a \triangleright_{\pi,P} b \quad \Longleftrightarrow \quad \exists i \in N \text{ such that } aP_ib \text{ and } \pi_i(b) > 0.$$

Lemma 4.4 extends the characterization of Bogomolnaia and Moulin (2001, Lemma 3) to multi-unit assignments. See Appendix A.3 for the proof.

Proposition 4.5. *For $n \geq 3$ and $m \geq 3$, the SRP mechanism is not SD-efficient.*

See Appendix A.4 for the proof.

Proposition 4.6 (Pairwise-dominance efficiency). *Every realization of SRP, unrestricted RSD, quota-RSD, the cyclic draft, and the snake draft is pairwise-dominance efficient.*

This is a standard efficiency property of sincere sequential picking (Brams and Straffin, 1979; Coreno and Balbuzanov, 2026).

See Appendix A.5 for the proof.

4.1 Ordinal Rank Balance

The new property, which we introduce in this paper, is that of Ordinal rank balance, which we define below.

Definition 4.7 (Ordinal rank balance). An allocation A satisfies *ordinal rank balance* if, for every agent i and every object $x \in A_i$,

$$|\{y \in A_i : yP_ix\}| \leq \sum_{j \neq i} |\{y \in A_j : yP_jx\}|.$$

Equivalently, for every $x \in A_i$, agent i holds at most half of all objects that are ranked above x by their own recipients.

In words, fix an object x received by agent i. Count the objects in i's bundle that she prefers to x. The other agents must collectively hold at least as many objects that they prefer to x, each judged by its recipient's ranking. Ordinal rank balance, therefore, bounds the inequality of the allocation in terms of ordinal preference concentration.

Proposition 4.8 (Ordinal rank balance). *Every realization of SRP, the cyclic draft, and the snake draft satisfies ordinal rank balance. In contrast, unrestricted RSD and quota-RSD with a common quota $k \geq 2$ do not satisfy ordinal rank balance in general.*

See Appendix A.6 for the proof.

With Proposition 4.8 we close the set of results which clarify the trade-offs presented by the mechanisms we consider and where SRP fits. Unrestricted RSD and quota-RSD protect incentives, but they allow early agents to accumulate highly ranked objects before others have a turn. Cyclic and snake drafts spread picking opportunities more evenly, yet their predictable turns create opportunities for manipulation. SRP retains the drafts' ordinal rank balance while making future turns uncertain enough to rule out misreports that yield a strict SD improvement.

5 The Two-Agent Case

This section considers $n = 2$. The initial picker is chosen uniformly, after which the picking sequence alternates deterministically: $1, 2, 1, 2, \ldots$ or $2, 1, 2, 1, \ldots$, each with probability $1/2$. Proposition 4.3 already covers this case, so SRP satisfies anonymity, equal treatment of equals, and weak SD-envy-freeness.

5.1 Properties under Stochastic Dominance

Proposition 5.1. *For two agents, the SRP mechanism fails weak SD-strategy-proofness.*

See Appendix A.7 for the proof.

Deterministic alternation allows an agent to manipulate her report by passing on an object that her opponent does not value highly, securing a contested object on her current turn, and receiving the deferred object at her guaranteed subsequent turn.

Theorem 5.2. *For two agents, the SRP mechanism is SD-efficient.*

See Appendix A.8 for the proof.

5.2 Ex-post Fairness

For the remainder of this section, assume that the number of objects m is even, so each agent receives exactly $m/2$ objects.

To analyze the fairness properties of this alternating structure, we introduce two relaxations of (ex-post) envy-freeness.

Definition 5.3 (Bidirectional Degree of Envy). For a deterministic allocation A and a preference profile P, the bidirectional degree of envy is the smallest non-negative integer k such that for every pair of distinct agents $i, j \in N$, there exist $G \subseteq A_i$ and $J \subseteq A_j$ with $|G| = |J| \leq k$, satisfying $(A_i \setminus G) R_i (A_j \setminus J)$. If no such k exists, the degree is $+\infty$. A mechanism is **k-bidirectional envy-free** if the bidirectional degree of envy of any produced allocation is at most k.

Definition 5.4 (Unidirectional Degree of Envy). For a deterministic allocation A and a preference profile P, the unidirectional degree of envy is the smallest non-negative integer k such that for every pair of distinct agents $i, j \in N$, there exists $J \subseteq A_j$ with $|J| \leq k$, such that $A_i R_i (A_j \setminus J)$. If no such k exists, the degree is $+\infty$. A mechanism is **k-unidirectional envy-free** if the unidirectional degree of envy of any produced allocation is at most k.

For the next proposition, we need to be explicit about how agents compare sets of objects. We will consider the case in which these preferences satisfy *responsiveness*.

Definition 5.5 (Responsive Preferences). Agent i's preferences are *responsive* if, for every bundle $S \subseteq O$ and every $x \in S$, $SP_i(S \setminus \{x\})$, and for every $y \in O \setminus S$,

$$(S \setminus \{x\}) \cup \{y\} \; P_i \; S \quad \iff \quad yP_ix.$$

Proposition 5.6. *For $n = 2$ and even m, if preferences are responsive, any realized allocation of the SRP mechanism is 1-bidirectional envy-free and 1-unidirectional envy-free.*

See Appendix A.9 for the proof.

6 Realized Inequality

Ex-ante symmetry alone does not reveal how unequal realized allocations can be. We therefore quantify inequality under SRP by first deriving the mean and variance of each agent's bundle size, and then calculating the expected variance of payoffs across agents under common object values drawn independently from Uniform$[0, 1]$. This complements the preceding fairness guarantees with explicit measures of dispersion in both the quantity and value of allocated objects.

Let $X_t = \mathbf{1}\{\sigma_t = i\}$ be the indicator that round t is assigned to agent i. The total number of objects allocated to agent i is $X = \sum_{t=1}^{m} X_t$.

Because the transition matrix T is symmetric and doubly stochastic, the stationary distribution of the Markov chain is uniform over all agents, $\mu = (\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n})$. Since the initial picker is chosen uniformly at random, the Markov chain is stationary from the first round, and the probability that agent i is the picker remains $\frac{1}{n}$ at every round t. Thus, $\mathbb{E}[X_t] = \Pr(X_t = 1) = \frac{1}{n}$ for all t, and the expected number of objects allocated to agent i is:

$$\mathbb{E}[X] = \sum_{t=1}^{m} \mathbb{E}[X_t] = \frac{m}{n}. \tag{6.1}$$

To compute the variance

$$\mathrm{Var}(X) = \sum_{t=1}^{m} \mathrm{Var}(X_t) + 2 \sum_{1 \leq t < s \leq m} \mathrm{Cov}(X_t, X_s),$$

we note that X_t is a Bernoulli($\frac{1}{n}$) random variable, yielding

$$\sum_{t=1}^{m} \mathrm{Var}(X_t) = m \cdot \frac{1}{n}\left(1 - \frac{1}{n}\right) = \frac{m(n-1)}{n^2}.$$

Due to the time-homogeneity of the Markov process, the covariance between any two rounds depends only on the time lag $d = s - t$. By lumping the process into a 2-state Markov chain with states $\{i\}$ and $N \setminus \{i\}$, the non-trivial eigenvalue is $\lambda = -\frac{1}{n-1}$. Thus, the d-step transition probability of returning to agent i is $T_{ii}^{(d)} = \frac{1}{n} + \frac{n-1}{n}\lambda^d$.

Thus, the covariance is:

$$\begin{aligned}
\mathrm{Cov}(X_t, X_{t+d}) &= \Pr(X_t = 1, X_{t+d} = 1) - \mathbb{E}[X_t]\mathbb{E}[X_{t+d}] \\
&= \frac{1}{n} T_{ii}^{(d)} - \frac{1}{n^2} \\
&= \frac{n-1}{n^2}\lambda^d.
\end{aligned}$$

Summing over all lags yields the variance of the number of items allocated to agent i:

$$\mathrm{Var}(X) = \frac{m(n-1)}{n^2} + \frac{2(n-1)}{n^2} \sum_{d=1}^{m-1} (m-d)\lambda^d. \tag{6.2}$$

Each agent therefore receives the same expected number of objects, m/n, although SRP does not impose equal realized bundle sizes.

While the bundle-size variance captures inequality in the quantity of allocated objects, we measure ex-post inequality in welfare using the variance of realized utilities. For a market with n agents and m objects, let Y_i denote the realized additive payoff of agent i under a given realization of the mechanism's randomization and agents' valuations, and let $\bar{Y} = \frac{1}{n}\sum_{j=1}^{n} Y_j$ be the average realized payoff across agents. We define expected ex-post inequality as

$$\mathbb{E}[V] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right], \tag{6.3}$$

where the expectation is taken over both the mechanism's randomization and the distribution of agents' valuations.

Evaluating this expression in closed form is challenging in general, since the set of available objects in each round depends on all agents' earlier assignments. As a benchmark, we consider the case where all agents share identical cardinal valuations, so the highest-value remaining object is assigned in each round. Specifically, let $u_i(o) = v(o)$ for every agent i and object o, where the common values $v(o)$ are drawn independently across objects from Uniform$[0, 1]$, independently of the mechanism's randomization.

6.1 Expected Payoff Inequality of SRP

To assess ex-post fairness, we establish the expected payoff variance of the allocated bundles.

Proposition 6.1. *Under uniform common values, the expected payoff variance for the SRP mechanism is given by:*

$$\mathbb{E}[V^{\mathrm{SRP}}] = \frac{n-1}{n^2}\left[\frac{m}{3} + \frac{1}{3(m+1)(m+2)}\sum_{d=1}^{m-1}(m-d)(m-d+1)(2m+d+4)\lambda^d\right],$$

where $\lambda = -\frac{1}{n-1}$.

See Appendix A.11 for the proof.

A Proofs and Supplementary Derivations

A.1 Proof of Theorem 4.2

Proof. Fix an agent i, a true preference profile P, and an arbitrary misreport P_i'. Let $\pi_i = \phi_i(P_i, P_{-i})$ and $\pi_i' = \phi_i(P_i', P_{-i})$ denote agent i's allocation vectors under truthful reporting and under the misreport, respectively.

For each sequence σ, let $A_i(\sigma)$ and $A_i'(\sigma)$ denote the bundles assigned to agent i under P_i and P_i', respectively. Suppose, to the contrary, that $\pi_i' R_i^{\text{sd}} \pi_i$ but $\pi_i' \neq \pi_i$. Thus, $\pi_i'(\text{Up}_i(o)) \geq \pi_i(\text{Up}_i(o))$ for all $o \in O$. Because $\pi_i' \neq \pi_i$, allocations must diverge on some positive-probability sequence. On any such sequence σ, let $t^* = t^*(\sigma)$ denote the earliest round at which the object assigned to agent i differs across the two reports. Because a divergence requires at least two available objects, $t^* \leq m-1$. Let S denote the set of objects assigned to agent i at round t^* under P_i across all diverging sequences and let x^* be the most preferred object in S according to P_i.

Fix any $w \in O$ with wP_ix^*. On any sequence with a first divergence at round t^*, the object x allocated to agent i at round t^* under P_i satisfies x^*R_ix, so wP_ix. If w were available at round t^*, agent i would receive w under P_i, so w must have been allocated prior to round t^*. Since allocations prior to round t^* are identical under both reports, w is allocated to the same agent under P_i and P_i'. On sequences where allocations never diverge, w is likewise allocated identically. Hence $\pi_i'(w) = \pi_i(w)$ for all wP_ix^*.

Turning to x^*, we show that $\mathbf{1}\{x^* \in A_i'(\sigma)\} \leq \mathbf{1}\{x^* \in A_i(\sigma)\}$ on every sequence σ. On sequences where allocations never diverge, equality is immediate. On a diverging sequence, suppose $x^* \in A_i'(\sigma)$. Then either agent i was allocated x^* before round t^* under P_i', in which case she also receives it under P_i because all allocations prior to round t^* are identical, or x^* was not allocated before round t^*. In the latter case, x^* remains available at round t^* under P_i, so agent i receives an object weakly preferred to x^*, which by definition of x^* must be x^* itself. Thus $x^* \in A_i(\sigma)$ whenever $x^* \in A_i'(\sigma)$ for all $\sigma \in \Omega$, establishing $\pi_i'(x^*) \leq \pi_i(x^*)$.

Moreover, this inequality holds strictly on a set of sequences with positive probability. Since $x^* \in S$, there exists a positive-probability sequence σ with first divergence at round t^* where agent i receives x^* under P_i and some $z \neq x^*$ under P_i'. Since $n \geq 3$, choose two distinct agents $j, k \neq i$. A continuation that alternates between j and k from round t^*+1 onward is feasible and has conditional probability $(n-1)^{-(m-t^*)} > 0$. Agent i is excluded from all remaining turns on this continuation, so x^* is allocated to another agent under P_i'. On these sequences, $\mathbf{1}\{x^* \in A_i(\sigma)\} = 1 > 0 = \mathbf{1}\{x^* \in A_i'(\sigma)\}$, yielding $\pi_i'(x^*) < \pi_i(x^*)$.

Therefore,

$$\begin{aligned}\pi_i'(\mathrm{Up}_i(x^*)) &= \sum_{wP_ix^*} \pi_i'(w) + \pi_i'(x^*) \\ &< \sum_{wP_ix^*} \pi_i(w) + \pi_i(x^*) \\ &= \pi_i(\mathrm{Up}_i(x^*)),\end{aligned}$$

contradicting the assumption that π_i' weakly stochastically dominates π_i. □

A.2 Proof of Proposition 4.3

Proof. Fix any two distinct agents $i, j \in N$. Because the initial picker is chosen uniformly at random and the Markov transition probabilities are symmetric across all agents, the distribution over picking sequences is invariant under any permutation of agents: for any permutation ρ of N, we have $\Pr(\rho(\sigma)) = \Pr(\sigma)$. Under the permuted profile P^ρ and sequence $\rho(\sigma)$, induction over rounds shows that the same object is selected at each round, with its recipient relabeled by ρ. Hence the mechanism is anonymous.

If $P_i = P_j$, and ρ is the transposition swapping i and j, then $\rho(P) = P$. By anonymity, $\phi(P) = \phi(\rho(P)) = \rho(\phi(P))$, which implies $\pi_i = \pi_j$. This establishes equal treatment of equals. It remains to establish weak SD-envy-freeness.

If agents i and j have identical rankings of individual objects, swapping their identities in the picking sequence swaps their bundles, since the mechanism uses only these rankings. Thus $\pi_i = \pi_j$, so strict SD-envy is impossible.

Now suppose their rankings of individual objects differ. Let k^* be the first rank at which these rankings diverge; necessarily $k^* \leq m - 1$. Let S be their common top set of size $k^* - 1$, with $S = \emptyset$ if $k^* = 1$, and let x and y be their respective k^*-th ranked objects, so that xP_iy and yP_jx. To rule out $\pi_j P_i^{\mathrm{sd}} \pi_i$, it suffices to show $\pi_i(\mathrm{Up}_i(x)) > \pi_j(\mathrm{Up}_i(x))$.

For any $o \in S$, both agents rank S identically at the top of their preferences, so neither receives an object outside S while objects in S remain available. Let ρ_{ij} be the transposition that swaps i and j in the picking sequence, leaving all other agents unchanged. Before S is exhausted, every round allocates the identical object under σ and $\rho_{ij}(\sigma)$, because agents i and j share the same ranking on S, while any other agent faces the same available set and preferences. Consequently, for every $o \in S$, we have $o \in A_i(\sigma) \iff o \in A_j(\rho_{ij}(\sigma))$. Since ρ_{ij} preserves sequence probabilities, $\pi_i(o) = \pi_j(o)$ for all $o \in S$, and hence $\pi_i(S) = \pi_j(S)$. Because $\mathrm{Up}_i(x) = S \cup \{x\}$ is a disjoint union,

$$\pi_i(\mathrm{Up}_i(x)) - \pi_j(\mathrm{Up}_i(x)) = \pi_i(x) - \pi_j(x).$$

We first show that $\pi_i(x) \geq \pi_j(x)$. Take any sequence σ in which j receives x, and consider the transposed sequence $\rho_{ij}(\sigma)$. Before S is exhausted, the two agents receive identical

objects whenever assigned a turn. Under σ, agent i cannot hold any turn between the exhaustion of S and the round where j receives x, since i would then receive x. Hence, at the earliest round after S is exhausted at which j holds the turn, i holds the turn in the transposed sequence. Since j receives x at or after this round in σ, object x has not yet been allocated to anyone, so it remains available and i receives x in the transposed sequence. Thus, $\rho_{ij}\Big(\{x \in A_j\}\Big) \subseteq \{x \in A_i\}$, and since transposition preserves sequence probabilities, $\pi_i(x) \geq \pi_j(x)$.

To see that the inequality is strict, let E^* be the set of sequences whose first k^*+1 rounds strictly alternate between i and j, with round k^* belonging to i. This fixes a feasible prefix with probability $\Pr(E^*) = 1/[n(n-1)^{k^*}] > 0$. In particular, when $n = 2$, E^* is one of the two alternating sequences and has probability $1/2$; no third agent is needed.

On E^*, the first $k^* - 1$ rounds exhaust S. At round k^*, agent i therefore receives x, her most-preferred remaining object. Under the transposed sequence $\rho_{ij}(\sigma)$, round k^* belongs to agent j. Because yP_jx and both x and y are available, agent j receives y. At round k^*+1, agent i holds the turn and receives x. Thus, j does not receive x under the transposed sequence. Hence $E^* \subseteq \{x \in A_i\} \setminus \rho_{ij}\Big(\{x \in A_j\}\Big)$. Because $\rho_{ij}\Big(\{x \in A_j\}\Big) \subseteq \{x \in A_i\}$ and $\Pr(E^*) > 0$, we obtain

$$\begin{aligned}\pi_i(x) &= \Pr(x \in A_i) \\ &> \Pr\Big(\rho_{ij}(\{x \in A_j\})\Big) \\ &= \pi_j(x).\end{aligned}$$

Therefore $\pi_i(\mathrm{Up}_i(x)) > \pi_j(\mathrm{Up}_i(x))$, ruling out $\pi_j P_i^{\mathrm{sd}} \pi_i$. By symmetry, $\pi_i P_j^{\mathrm{sd}} \pi_j$ cannot hold either, so no agent strictly SD-envies another, and thus the mechanism is weakly SD-envy-free. □

A.3 Proof of Lemma 4.4

Proof. **Only if statement:** Suppose $\triangleright_{\pi,P}$ contains a cycle:

$$a_1 \triangleright_{\pi,P} a_2 \triangleright_{\pi,P} \ldots \triangleright_{\pi,P} a_K \triangleright_{\pi,P} a_1,$$

where $a_1, \ldots, a_K$ are distinct. By definition of $\triangleright_{\pi,P}$, for each $k \in \{1, \ldots, K\}$, there exists an agent i_k such that $a_k P_{i_k} a_{k+1}$ and $\pi_{i_k}(a_{k+1}) > 0$ with $a_{K+1} = a_1$.

Choose ϵ such that $0 < \epsilon \leq \min_{1 \leq k \leq K} \pi_{i_k}(a_{k+1})$, and define π' by shifting ϵ probability from a_{k+1} to a_k for each agent i_k. By construction, π' is a feasible allocation. Each transfer moves probability from a less-preferred object a_{k+1} to a strictly preferred object a_k, so the resulting allocation satisfies $\pi'_i P_i^{\mathrm{sd}} \pi_i$ for every participating agent i. Uninvolved agents receive

unchanged assignment vectors. Thus, $\pi_i' R_i^{\text{sd}} \pi_i$ for every agent, with $\pi_i' P_i^{\text{sd}} \pi_i$ for at least one agent, so π is not SD-efficient.

If statement: Suppose π is not SD-efficient, so there exists a feasible allocation π' such that $\pi_i' R_i^{\text{sd}} \pi_i$ for every agent i, and $\pi_{i_1}' P_{i_1}^{\text{sd}} \pi_{i_1}$ for at least one agent i_1. Thus, every agent weakly improves under SD, and at least one strictly improves. By stochastic dominance, there exist objects a_1, a_2 such that $\pi_{i_1}'(a_1) < \pi_{i_1}(a_1), \quad \pi_{i_1}'(a_2) > \pi_{i_1}(a_2)$, and $a_2 P_{i_1} a_1$. Since $\pi_{i_1}(a_1) > 0$, this gives $a_2 \rhd_{\pi,P} a_1$.

Since $\sum_{i \in N} \pi_i'(a_2) = \sum_{i \in N} \pi_i(a_2) = 1$, the inequality $\pi_{i_1}'(a_2) > \pi_{i_1}(a_2)$ implies that some agent i_2 must receive strictly less of a_2, so $\pi_{i_2}'(a_2) < \pi_{i_2}(a_2)$. Because $\pi_{i_2}' R_{i_2}^{\text{sd}} \pi_{i_2}$, there must exist an object a_3 such that $a_3 P_{i_2} a_2$ and $\pi_{i_2}'(a_3) > \pi_{i_2}(a_3)$. Since $\pi_{i_2}(a_2) > 0$, this gives $a_3 \rhd_{\pi,P} a_2$.

Continuing this process yields an infinite sequence $a_1, a_2, a_3, \ldots$ with $a_{k+1} \rhd_{\pi,P} a_k$ for each $k \geq 1$. Because O is finite, some object must eventually repeat, which forms a cycle in $\rhd_{\pi,P}$. □

A.4 Proof of Proposition 4.5

Proof. By Lemma 4.4, it suffices to exhibit a preference profile whose induced allocation contains a cycle in $\rhd_{\pi,P}$.

Fix three distinct objects $a, b, c \in O$. Suppose agents 1, 2, and 3 rank these items at the top according to $aP_1bP_1cP_1 \ldots$, $bP_2aP_2cP_2 \ldots$, and $cP_3aP_3bP_3 \ldots$, with any other agents' preferences and all remaining rankings arbitrary.

The Markov chain switches agents at every round. A turn sequence beginning with $(2, 3, 2)$ occurs with positive probability, under which agent 2 receives b, agent 3 receives c, and agent 2 subsequently receives a, hence $\pi_2(a) > 0$. Similarly, any sequence starting with $(1, 3, 1)$ occurs with positive probability and allocates a and b to agent 1, so $\pi_1(b) > 0$. Combined with aP_1b and bP_2a, this implies $a \rhd_{\pi,P} b$ and $b \rhd_{\pi,P} a$. Thus $\rhd_{\pi,P}$ is not acyclic, and the mechanism is not SD-efficient. □

A.5 Proof of Proposition 4.6

Proof. Fix a realization of any of the five mechanisms, and let A be the resulting allocation. For unrestricted RSD, regard the first agent as selecting all objects in order of preference. Write $x_1, \ldots, x_m$ for the objects in their order of selection, and let i_t be the agent receiving x_t. In every case, each turn assigns the highest-ranked available object. The argument below uses only this rule.

Suppose, toward a contradiction, that some feasible allocation $B \neq A$ pairwise-dominates A. Let t be the first position at which x_t has a different recipient in B, and set $i = i_t$.

Since agent i selected her most-preferred available object at position t, every object she prefers to x_t was selected before t. By the choice of t, all these earlier objects have the same recipients in A and B.

Thus, using the upper-contour set $\mathrm{Up}_i(x_t)$, we have

$$|B_i \cap \mathrm{Up}_i(x_t)| = |A_i \cap \mathrm{Up}_i(x_t)| - 1,$$

because i receives x_t in A but not in B.

However, pairwise dominance requires an injection $f_i : A_i \to B_i$ that maps every object to a weakly preferred replacement. Such an injection must map the distinct objects in $A_i \cap \mathrm{Up}_i(x_t)$ to distinct objects in $B_i \cap \mathrm{Up}_i(x_t)$, contradicting the inequality above.

Therefore, no feasible allocation pairwise-dominates A. □

A.6 Proof of Proposition 4.8

Proof. Fix a realized picking sequence and its allocation A. Consider any agent i and object $x \in A_i$. Suppose that i receives x on her r-th turn, which occurs at position t in the picking sequence.

Since agents always select their highest-ranked available object, the objects in A_i ranked above x are exactly those selected by i before position t. Thus,

$$|\{y \in A_i : yP_ix\}| = r - 1.$$

Moreover, every object y selected before position t by another agent $j \neq i$ satisfies yP_jx, since x was still available when j selected y. There are $t - r$ such objects, so

$$\sum_{j \neq i} |\{y \in A_j : yP_jx\}| \geq t - r.$$

Consequently, ordinal rank balance holds whenever $t - r \geq r - 1$.

Under SRP, consecutive turns belong to different agents. Hence, between any two consecutive selections by i, at least one other agent selects an object. Therefore, $t - r \geq r - 1$.

Under both the cyclic and snake drafts, each complete round of n turns assigns exactly one turn to each agent. Before i makes her r-th selection, at least $r - 1$ complete rounds have elapsed, during which the other agents collectively made $(n - 1)(r - 1)$ selections. Thus,

$$t - r \geq (n - 1)(r - 1) \geq r - 1,$$

where the last inequality follows from $n \geq 2$. Therefore, all three mechanisms satisfy ordinal rank balance in every realization.

To establish the negative results, consider a profile in which all agents have the identical strict ranking

$$P_1 = \cdots = P_n : \quad o_1, o_2, \ldots, o_m.$$

Under unrestricted RSD, the first agent receives every object. Under quota-RSD with $k \geq 2$, the first agent receives the top k objects. In either case, letting i denote the first agent and $x = o_2$, we have

$$|\{y \in A_i : yP_ix\}| = 1, \qquad \sum_{j \neq i} |\{y \in A_j : yP_jx\}| = 0.$$

Hence ordinal rank balance is violated. □

A.7 Proof of Proposition 5.1

Proof. Consider two agents $N = \{1, 2\}$ and four objects $O = \{a, b, c, d\}$, with true preferences $P_1 : c, d, b, a$ and $P_2 : d, b, c, a$.

Under truthful reporting, the sequences $1, 2, 1, 2$ and $2, 1, 2, 1$ each occur with probability $\frac{1}{2}$, allocating bundles $\{b, c\}$ and $\{a, c\}$ to agent 1, respectively. The resulting allocation vector assigns probabilities 1 to c, $\frac{1}{2}$ to b, $\frac{1}{2}$ to a, and 0 to d.

Now suppose agent 1 misreports $P_1' : d, c, b, a$. The same two sequences allocate bundles $\{c, d\}$ and $\{a, c\}$ to agent 1, assigning probabilities 1 to c, $\frac{1}{2}$ to d, $\frac{1}{2}$ to a, and 0 to b.

The misreport shifts probability $\frac{1}{2}$ from b to the strictly preferred object d while leaving all other probabilities unchanged. Hence $\pi_1' P_1^{\mathrm{sd}} \pi_1$, showing that the mechanism fails weak SD-strategy-proofness. □

A.8 Proof of Theorem 5.2

Proof. By Lemma 4.4, it suffices to show that $\triangleright_{\pi,P}$ is acyclic.

The turn sequence alternates strictly between the two agents. The only two possible sequences are $\sigma^1 = (1, 2, 1, 2, \ldots)$ and $\sigma^2 = (2, 1, 2, 1, \ldots)$, each occurring with probability $1/2$.

We first show that any cycle in $\triangleright_{\pi,P}$ implies a cycle of length two. In any minimal directed cycle, two consecutive edges cannot belong to the same agent, because $a \xrightarrow{i} b \xrightarrow{i} c$ implies aP_ic by transitivity and $\pi_i(c) > 0$, meaning the edge $a \xrightarrow{i} c$ is also present, which contradicts minimality. Hence, agents must alternate along any minimal cycle. If a minimal cycle has length four or more, say $a_1 \xrightarrow{1} a_2 \xrightarrow{2} a_3 \xrightarrow{1} a_4$, then either $a_1P_1a_4$ (yielding the edge $a_1 \xrightarrow{1} a_4$ and shortening the cycle) or $a_4P_1a_1$, in which case transitivity gives $a_3P_1a_2$ and $\pi_1(a_2) > 0$, forming a 2-cycle with $a_2 \xrightarrow{2} a_3$. Thus, $\triangleright_{\pi,P}$ has a cycle if and only if there exist distinct objects $x, y \in O$ satisfying xP_1y, yP_2x, $\pi_1(y) > 0$, and $\pi_2(x) > 0$.

We prove by induction on $m = |O|$ that no such pair (x, y) exists. Let $a = \max_{P_1} O$ and $b = \max_{P_2} O$. If $a = b$, object a cannot belong to $\{x, y\}$ because xP_1y rules out $y = a$ and yP_2x rules out $x = a$. If $a \neq b$, each agent receives their top item with certainty in the first two rounds, ruling out $x = a$ and $y = b$, while yP_2x and xP_1y rule out $x = b$ and $y = a$. Thus, any such pair must satisfy $\{x, y\} \subseteq O \setminus \{a, b\}$.

For the base cases, no pair of distinct objects exists when $m = 1$. When $m = 2$, if the agents share a top object, their rankings coincide and no 2-cycle is possible; if their top objects differ, each receives her top object under both sequences, again ruling out a 2-cycle. For $m = 3$, if $a \neq b$, then $|O \setminus \{a, b\}| = 1$, which cannot contain two distinct objects. If $a = b$, round 1 allocates a under both sequences, so the remaining set is exactly $\{x, y\}$. Because xP_1y and yP_2x, agent 1 receives x and agent 2 receives y under both sequences, yielding $\pi_1(y) = \pi_2(x) = 0$. Thus no cycle exists for $m = 3$.

For the inductive step with $m \geq 4$, assume no cycle exists in any problem with fewer than m objects. Suppose towards contradiction that a pair (x, y) exists for m objects. If $a = b$, round 1 allocates a under both sequences. If $a \neq b$, the first two rounds allocate a to agent 1 and b to agent 2 under both sequences. In either case, the remaining process is the alternating mechanism on the remaining objects (with cardinality $m - 1$ or $m - 2 \geq 2$), with the two possible continuations each occurring with probability $\frac{1}{2}$. Thus, the pair (x, y) would still form a cycle in a problem with strictly fewer than m objects, contradicting the induction hypothesis. Hence $\triangleright_{\pi,P}$ is acyclic, and the mechanism is SD-efficient. □

A.9 Proof of Proposition 5.6

Proof. Write $q = m/2$, and suppose without loss of generality that agent 1 picks first. Let $o_{i,k}$ be the object assigned to agent i on her k-th turn, so $A_i = \{o_{i,1}, \ldots, o_{i,q}\}$.

When agent 1 receives $o_{1,k}$, object $o_{2,k}$ is still available. Thus $o_{1,k}P_1o_{2,k}$ for every $k = 1, \ldots, q$. Repeatedly replacing each $o_{2,k}$ by $o_{1,k}$, responsiveness and transitivity give $A_1P_1A_2$. Applying the same replacements for $k = 1, \ldots, q - 1$ gives

$$A_1 \setminus \{o_{1,q}\} \; R_1 \; A_2 \setminus \{o_{2,q}\}.$$

When agent 2 receives $o_{2,k}$, object $o_{1,k+1}$ is still available. Hence $o_{2,k}P_2o_{1,k+1}$ for $k = 1, \ldots, q-1$, and the same replacement argument gives

$$A_2 \setminus \{o_{2,q}\} \; R_2 \; A_1 \setminus \{o_{1,1}\}.$$

For $q = 1$, both displayed comparisons hold because the two reduced bundles are empty. These comparisons establish 1-bidirectional envy-freeness for both agents.

Finally, responsiveness gives $A_i P_i(A_i \setminus \{o_{i,q}\})$. Combining this with the displayed comparisons yields $A_1 P_1(A_2 \setminus \{o_{2,q}\})$ and $A_2 P_2(A_1 \setminus \{o_{1,1}\})$, establishing 1-unidirectional envy-freeness. □

A.10 The General Expression for Expected Payoff Variance

Let $A = (A_i)_{i \in N}$ denote the realized allocation, where A_i is agent i's bundle. We assume agents have additive utilities over bundles, with finite second moments for all object valuations. The realized payoff for agent i is:

$$Y_i = \sum_{o \in A_i} u_i(o),$$

where $u_i(o)$ denotes agent i's cardinal valuation for object o. The variance across agents within each realized allocation is:

$$V = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \bar{Y})^2, \qquad \bar{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

Let o_t be the object allocated in round t, and let $\sigma_t \in \{1, \ldots, n\}$ denote the agent who picks in round t. We define:

$$W_t = u_{\sigma_t}(o_t), \qquad H_{ts} = \mathbf{1}\{\sigma_t = \sigma_s\}, \qquad \lambda = -\frac{1}{n-1}.$$

Here, W_t is the realized payoff in round t and H_{ts} indicates whether the same agent picks in both rounds t and s. The Markov-chain transition of the mechanism implies:

$$\Pr(H_{ts} = 1) = \frac{1}{n} + \frac{n-1}{n} \lambda^{|s-t|}.$$

Because $Y_i = \sum_{t=1}^{m} \mathbf{1}\{\sigma_t = i\} W_t$ and $\sum_{i=1}^{n} \mathbf{1}\{\sigma_t = i\} = 1$, we have:

$$\sum_{i=1}^{n} Y_i^2 = \sum_{t=1}^{m} W_t^2 + 2 \sum_{t<s} H_{ts} W_t W_s,$$
$$\sum_{i=1}^{n} Y_i = \sum_{t=1}^{m} W_t.$$

Consequently, the variance expands to:

$$V = \frac{n-1}{n^2} \sum_{t=1}^{m} W_t^2 + \frac{2}{n^2} \sum_{t<s} (n H_{ts} - 1) W_t W_s.$$

Since $\mathbb{E}[nH_{ts} - 1] = n\Pr(H_{ts} = 1) - 1 = (n-1)\lambda^{s-t}$, the expectation of the cross-term expands as:

$$\begin{aligned}\mathbb{E}[(nH_{ts} - 1)W_t W_s] = (n-1)\lambda^{s-t}\mathbb{E}[W_t W_s] \\ + n\,\mathrm{Cov}(H_{ts}, W_t W_s).\end{aligned}$$

Substituting this back into the expectation of V and letting $d = s - t$ yields the general expression for expected payoff variance:

$$\begin{aligned}\mathbb{E}[V^{\mathrm{SRP}}] = \frac{n-1}{n^2}\Bigg[\sum_{t=1}^{m}\mathbb{E}[W_t^2] \\ + 2\sum_{d=1}^{m-1}\lambda^d\sum_{t=1}^{m-d}\mathbb{E}[W_t W_{t+d}]\Bigg] \\ + \frac{2}{n}\sum_{d=1}^{m-1}\sum_{t=1}^{m-d}\mathrm{Cov}(H_{t,t+d}, W_t W_{t+d}).\end{aligned} \tag{A.1}$$

The additional covariance term captures the path-dependence of the available objects on the history of picker identities.

A.11 Proof of Proposition 6.1

Proof. Under identical cardinal valuations, let $v_1 \geq v_2 \geq \cdots \geq v_m$ be the common values in descending order. Regardless of the picking sequence, the object allocated in any round t is deterministically the t-th best object, yielding $W_t = v_t$. Hence, the allocated values W_t and W_{t+d} depend only on the common values and are independent of the picking sequence, implying $\mathrm{Cov}(H_{t,t+d}, W_t W_{t+d}) = 0$.

Equation (A.1) then simplifies to:

$$\mathbb{E}[V^{\mathrm{SRP}}] = \frac{n-1}{n^2}\left[\sum_{t=1}^{m}\mathbb{E}[v_t^2] + 2\sum_{d=1}^{m-1}\lambda^d \sum_{t=1}^{m-d}\mathbb{E}[v_t v_{t+d}]\right]. \tag{A.2}$$

Since v_t are descending order statistics of m independent uniform draws, their moments satisfy:

$$\mathbb{E}[v_t^2] = \frac{(m-t+1)(m-t+2)}{(m+1)(m+2)},$$
$$\mathbb{E}[v_t v_{t+d}] = \frac{(m-t-d+1)(m-t+2)}{(m+1)(m+2)}.$$

Summing these moments yields:

$$\sum_{t=1}^{m}\mathbb{E}[v_t^2] = \frac{m}{3},$$
$$\sum_{t=1}^{m-d}\mathbb{E}[v_t v_{t+d}] = \frac{(m-d)(m-d+1)(2m+d+4)}{6(m+1)(m+2)}.$$

Substituting into (A.2) yields:

$$\mathbb{E}[V^{\mathrm{SRP}}] = \frac{n-1}{n^2}\left[\frac{m}{3} + \frac{1}{3(m+1)(m+2)}\sum_{d=1}^{m-1}(m-d)(m-d+1)(2m+d+4)\lambda^d\right]. \qquad \square$$

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